



FPVspeed™

DIY Module Series

FPV Video Receiver

FS3138RX

3100~3800MHz -95dBm SPI & RSSI

1. 通用规格 (General)

FS3138RX	
Weight	5.5g±0.1g
Dimensions	L37 x W26.3 x H4.5mm
DC Characteristics	
Power Supply	5.0V
Current Consumption	420mA ±30mA @5V
Environmental Specification	
Operating Temperature	-10~+65°C
Storing Temperature	-30~+85°C
Operating Humidity	85%RH
RF	
Receiving Frequency Range	3100~3800MHz
Demodulation System	FM/PLL
IF	480MHz
ANT. Input Impedance	50Ω, Typ.
LO Frequency Stabilization	±200kHz
LO Frequency Precision	±200kHz
LO Control	PLL
Input LO Leak	-65dBm
Receiving Sensitivity	-95dBm±3dBm
Video Characteristics	
Video Output Impedance	75Ω, Typ.
Video Output Level	1±0.2Vp-p, Typ.
Video Polarity	Negative
Video Frequency Response	±5dB, Max. 50Hz~6MHz
Differential Gain	±5%, Max
Differential Phase	±5Deg., Max
3Db IF WIDEBAND	16.5MHz
S/N	38dB, Min
RSSI Slope	22mv/dBm



APPLICATIONS

- FPV Drone Racing
- UAV Communication Systems
- Remote Vehicle Monitoring
- Wireless Video Transmission Systems

• Code change steps for running V1.2 50MHz code on your V1.5 20MHz crystal board

- ① Open file PLL.h
- ② Change #define ref 50 → #define ref 20, keep #define fref_div 1 as original.
- ③ No edits for any .c file (SPI, frequency table, PLL calculation code all same).
- ④ Compile directly, RF output frequency keeps original range.



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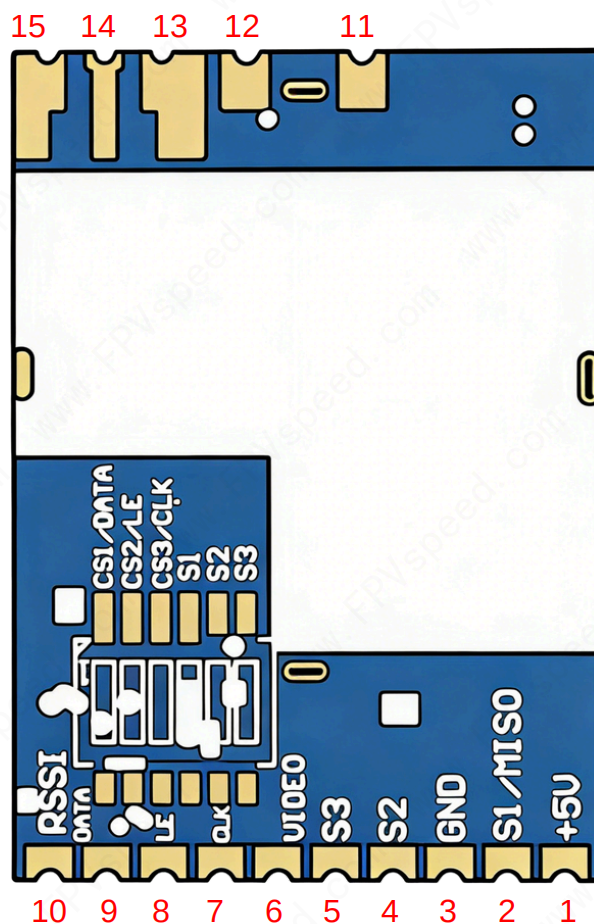
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2.引脚分布 (Pin Assignment)

Pin No.	Function
1	+5V IN
2	S1: NC
3	GND
4	S2: NC
5	S3: NC
6	Video Out
7	SPI_CLK / CS3
8	SPI_LE / CS2
9	SPI_DATA / CS1
10	RSSI
11	GND
12	GND
13	GND
14	ANT
15	GND

PAD No.	Function
2	Connected with S1
4	Connected with S2
5	Connected with S3
7	Connected with CS3
8	Connected with CS2
9	Connected with CS1



Using the E4421B signal generator as the signal input source for the module, the RSSI output voltage at the 3170MHz frequency point is for reference only.

Input Signal Strength	RSSI Output Voltage	Input Signal Strength	RSSI Output Voltage
-136dB	0.66V	-65dBm	2.40V
-95dB	0.81V	-60dBm	2.68V
-90dB	1.01V	-55dBm	2.81V
-85dB	1.28V	-50dBm	2.81V
-80dB	1.56V	-45dBm	2.83V
-75dB	1.84V	-40dBm	2.83V
-70dBm	2.11V		



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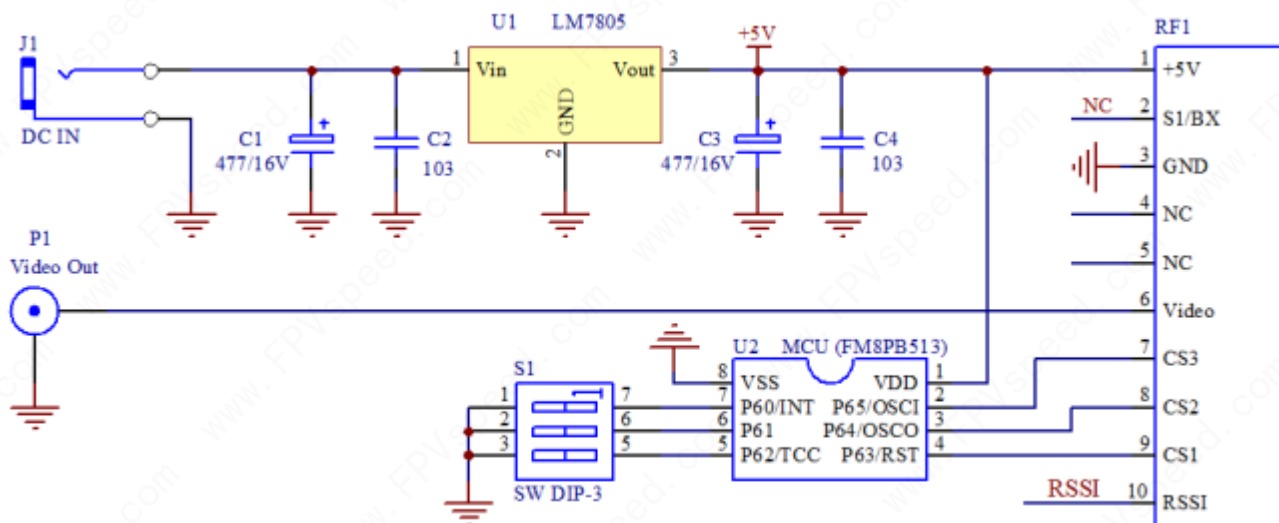
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3.应用电路 (Application Circuit)



4.结构尺寸 (Dimension)

